

Lab 3

Uniform Cost Search

Imagine that a person wants to travel from place A to place D. There are several roads between these places, and each road has different cost. Some roads are cheap, while some roads are expensive. The person does not simply want to reach D as quickly as possible. The main aim is to reach D by spending the least amount of money.

The journey starts at A. From A, there are two possible roads. One road goes from A to B and costs 1. Another road goes from A to C and costs 4. At this point, the person has two choices. Going directly to C costs 4, while going to B costs only 1. So the journey through B is cheaper at the moment.

From B, there are again two choices. One road goes directly from B to D and costs 5. Another road goes from B to C and costs 2. The person does not simply choose the road that reaches D first. Instead, the total money spent so far is considered. Going from A to B costs 1, and then going from B to D, costs 5. Therefore the total cost of $A \rightarrow B \rightarrow D$ is $1+5=6$.

The other choice from B is go to C. The journey from A $\rightarrow$ B costs 1, and the journey from B $\rightarrow$ C costs 2. Therefore, the total cost of $A \rightarrow B \rightarrow C$ is $1+2=3$. From C, there is a road to D that costs 1. So the complete journey $A \rightarrow B \rightarrow C \rightarrow D$ costs $1+2+1=4$.

There is also another possible journey. The person can directly go from $A \rightarrow C$, which costs 4, and then travel from $C \rightarrow D$, which costs 1. Therefore, the total cost of $A \rightarrow C \rightarrow D$ is $4 + 1 = 5$.

Now, there are 3 possible ways to reach D. The first way is $A \rightarrow B \rightarrow D$, which costs 6. The second way is $A \rightarrow C \rightarrow D$, which costs 5. The third way is $A \rightarrow B \rightarrow C \rightarrow D$, which costs only 4. When the person compares all the journeys, the cheapest one is clearly $A \rightarrow B \rightarrow C \rightarrow D$, with a total cost of 4.

This is the main idea of Uniform Cost Search. Imagine a person standing at the crossroads, with several possible journeys written on pieces of paper. Each piece of paper shows how much money has been spent so far and where the person has travelled. The person always looks at the journey that has cost the least amount of money so far and continues from there. If a new journey is created, its total cost is calculated and compared with the other journeys.

The important thing is that the person does not choose a journey just because it has fewer roads. For example, $A \rightarrow B \rightarrow D$ has only two roads, but it costs 6. On the other hand, $A \rightarrow B \rightarrow C \rightarrow D$ has three roads, but it costs only 4. Therefore, the three road journey is better because it requires less money.

We can understand this with a simple real-life

example. Suppose you want to travel to your college. One route may only have two roads, but there may be expensive tolls on the way. Another route may have four roads but no expensive tolls. If your main goal is to spend less money, you would choose the second route even though it has more roads. UCS works in exactly this way. It cares about the total cost, not the number of steps.

Another example is a delivery person carrying a parcel. There may be many different roads to reach the customer's house. One road may be very short but have a high toll. Another road may be longer but have a very small toll. The deliver person can compare the total travel cost of different routes and choose the route that costs the least. This makes the journey more economical.

The same idea can also be used by a robot moving from one room to another. Suppose some paths are easy for the robot to travel, while other paths require more energy. The robot should not simply choose the path with the fewest movements. Instead, it should choose the path that requires the least total energy. In the same way, UCS chooses the journey with the smallest total cost.

In this example, the cheapest journey is $A \rightarrow B \rightarrow C \rightarrow D$ and its total cost is 4. Therefore, the final answer is

a cost of 4 and the path A, B, C, D.

The main lesson from this experiment is that reaching the destination first does not always mean that we have found the best route. A route with fewer steps can be more expensive, while a route with more steps can be cheaper. UC carefully compares the total cost of different choices and selects the least expensive route. Therefore, it is useful whenever the main goal is to find a path with the minimum total cost, such as finding cheaper travel routes, reducing delivery costs, saving fuel, or reducing the energy used by a robot.